

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code: ITA03

CO Assessed: CO1

Weightage:

CO1 Statement:

Student Name:

Section / Batch:

30 % of CIA

Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.

G. Mythili
S101D

Course Title:

Assessment:

Total Marks:

Mobile computing

Assessment Tool 2 – Short Answer

100Marks

Reg. No.:

Date:

192372250

28/07/26

Code of Conduct

I G. Mythili, 192372250 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Mythili

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

90
100

A total spectrum = 45 MHz = 45000 kHz
 full duplex needs 2 simple channels = 2 x 25 = 50 kHz
 channels = 45000 / 50 = 900
 Ans: 900

2) channels per cell = total channels / N = 560 / 7 = 80

3) Total voice channels = carriers x slots / carrier
 = 100 x 8 = 800

4) power ratio = 80 / 4 = 20
 Ratio in dB = $10 \times \log_{10}(20) = 13.01 \text{ dB} = 13 \text{ dB}$

5) $CI(\text{dB}) = \text{Signal}(\text{dBm}) - \text{interference}(\text{dBm}) = -60 - (-92) = 32 \text{ dB}$

6) Total channels = 42000 kHz / 300 kHz = 140
 channels per cell = 140 / 7 = 20 Ans: 20

7) Received power $\propto 1/d^n$ Doubling distance with $n=4$ reduce power
 by $2^4 = 16 \times$ new power = $64 / 16 = 4 \text{ W}$ Ans: 4W

8) with $n=2$, doubling radius requires power increase by
 $2^2 = 4 \times$ to maintain same edge SNR
 Ans: 4x

9) Total channels in cluster = cluster size x channels / cell = $4 \times 15 = 60$

10) $V = 100 \text{ km/h} = 27.78 \text{ m/s}$

Doppler shift = $V \times f / c = 27.78 \times 900 \times 10^6 / 3 \times 10^8$
 = 83.34 Hz = 83 Hz

Ans = 83 Hz

11) slot duration = frame time / slots = $\frac{5 \text{ ms}}{7.68} = 0.65 \text{ ms}$
 Ans: 0.65 ms

12) Carriers: Total spectrum / spacing = $\frac{24000 \text{ kHz}}{200 \text{ kHz}} = 120$
 Ans: 120

13) Tx power in dBm = $10 \log_{10}(200 \text{ mW}) = 39.03 \text{ dBm}$
 Received power = $39.03 - 33 = 6.03 \text{ dBm}$
 Ans: 6 dBm

14) using the log distance path model referenced to 1m
 Scaling the distance to 2.5 km gives an approximate
 free space path loss of = 123 dB
 Ans: 123 dB

15) Total capacity = calls/cell $\times$ cells = $250 \times 6 = 1500$
 Ans: 1500

16) Active calls supported = arrival rate $\times$ avg duration = $4000 \text{ calls/sec} \times 180$
 Ans: 720000

17) total bandwidth = channels $\times$ bandwidth/channel = $150 \times 25 \text{ kHz}$
 = 3.75 MHz
 Ans: 3.75 MHz

18) Number of guard bands between 30 bands = 29 wasted
 bandwidth = $29 \times 40 \text{ kHz} = 1160 \text{ kHz} = 1.16 \text{ MHz}$
 Ans: 1.16 MHz

19) Speed = 90 km/h = 25 m/s cell diameter = $2 \times 1.5 \text{ km} = 3000 \text{ m}$ time per
 cell = $3000 / 25 = 120 \text{ s}$ Handoffs / hour = $3600 / 120 = 30$
 Ans: 30

all users = regional centers & base stations / content users / base stations

$$= 8 \times 10 \times 250 = 80,000$$

Ans: 80,000

Approximate cell area: $P = (0.4 \text{ km})^2 = 0.16 \text{ km}^2$

Number of cells = $400 / 0.16 = 2500$

Ans: 2500

channels = Total spectrum / channel width = $6000 \text{ kHz} / 20 \text{ kHz} = 200$

Ans: 200

8) Total call-minutes = $1200 \times 4 \times 2.5 = 12,000 \text{ min/hr}$ Traffic in

Ans: $12,000 / 60 = 200$

Ans: 200

24) Users = Total bandwidth / bandwidth per user = $40,000 \text{ kHz} / 400 \text{ kHz} = 100$

Ans: 100

25) Total capacity = users / cell $\times$ cells = $60 \times 1200 = 72000$

Ans: 72000

26) paging operations = Total cells / cells per operation = $200 / 4 = 50$

27) offered traffic = arrival rate $\times$ avg duration = $40000 \times 3 = 120000$

Erlangs

Excess over capacity = $120000 - 100000 = 20,000$

fraction blocked = $20000 / 120000 = 0.17$ Ans: 0.17

28) Roaming users = $15\% \times 600000 = 90,000$

Ans: 90,000

29) utilization - peak traffic / processing rate = $9000 / 12000 = 0.75 = 75\%$

Ans: 75%

30) Total channels = Bandwidth / channel width = $48000 / 200 = 240$ channels

per cell = $240 / 4 = 60$

Ans: 60

31) channels = Total spectrum / channel width = $30,000 / 200 = 150$

32) successful hand-offs = $99\% \times 2500 = 2475$
Ans: 2475

33) Tx power in dBm = $10 \log_{10} (12000 \text{ mW}) = 40.79 \text{ dBm}$

Received power = $40.79 - 45 = -4.21 \text{ dBm} = -4 \text{ dBm}$ Ans: -4 dBm

34) Total Capacity = users/cell $\times$ cells = $900 \times 7 = 6300$
Ans: 6300

35) slot duration = frame time / slots = $4.8 \text{ ms} / 8 = 0.6 \text{ ms}$
Ans: 0.6 ms

36) $V = 150 \text{ km/h} = 41.67 \text{ m/s}$. Doppler shift = $V \times f / c = \frac{41.67 \times 1.8}{3 \times 10^8}$
 $= 250 \text{ Hz}$ Ans: 250 Hz

37) offered traffic = $70,000 \times 4 = 280,000 \text{ Erlangs}$

Excess = $280,000 - 250,000 = 30,000$

fraction blocked = $30,000 / 280,000 = 0.11$ Ans: 0.11

38) Total users in cluster = users/cell $\times$ cells in cluster = $600 \times 9 = 5400$
Ans: 5400

39) calls dropped/sec = $0.4\% \times 60,000 = 240$
Ans: 240

40) path loss (dB) = Tx power (dBm) - Rx power (dBm) = $43 - (-87) = 130 \text{ dB}$
Ans: 130 dB

41) Total coverage = coverage / station $\times$ number of stations
 $= 2.5 \times 150 = 375 \text{ km}^2$
Ans: 375 km²

42) Total channels = $48,000 / 200 = 240$
channels per cell = $240 / 6 = 40$
Ans: 40

$$\text{paging cycles} = \text{Total users} / \text{paging cycle} \\ = 150,000 / 6000 = 25$$

Ans: 25

$$\text{Handoffs per cell} = \text{call duration} / \text{handoff interval} \\ = 9 / 3 = 3$$

Ans: 3

$$\text{SNR(dB)} = \text{Signal(dBm)} - \text{Noise(dBm)} \\ = -85 - (-115) = 30 \text{ dB}$$

$$\text{Minimum bandwidth} = \text{Users} \times \text{rate/user} \\ = 250 \times 150 \text{ kbps} = 37,500 \text{ kbps} = 37.5 \text{ Mbps}$$

Ans: 37.5 Mbps

$$\text{Total call-minutes/hr} = \text{Users} \times \text{calls/hr} \times \text{avg duration} \\ = 1500 \times 4 \times 2.5 = 15,000 \text{ min}$$

$$\text{Traffic in Erlangs} = 15,000 / 60 = 250$$

$$\text{Total bandwidth} = \text{Channels} \times \text{bandwidth/channel} = 250 \times 12 \text{ kHz} \\ = 3000 \text{ kHz} \\ = 3 \text{ MHz} \text{ Ans: 3 MHz}$$

$$\text{Total throughput} = \text{Connections} \times \text{rate/connection} = 1500 \times 600 \text{ kbps} \\ = 900,000 \text{ kbps} \\ = 900 \text{ Mbps} \text{ Ans: 900 Mbps}$$

$$\text{Area per tower} = \pi \times r^2 = 50.27 \text{ km}^2 \text{ Towers needed} = 350 / 50.27 \\ = 6.96 = 7$$

Ans: 7